

Multiple Choice Questions (MCQs)

1. What is a necessary condition for a deadlock to occur?

- A) Resource sharing
- B) Mutual Exclusion
- C) Resource preemption
- D) Process termination

Answer: B) Mutual Exclusion

2. Which of the following is an example of the Hold and Wait condition?

- A) A process is waiting for a resource that it does not hold.
- B) A process holds a file and waits for a printer.
- C) A process has released all its resources.
- D) A process is terminated.

Answer: B) A process holds a file and waits for a printer.

3. In the context of deadlock, what does the No Preemption condition imply?

- A) Resources can be forcibly taken from a process.
- B) Resources must be released voluntarily by the process.
- C) Processes can be terminated to free resources.
- D) Resources can be shared among processes.

Answer: B) Resources must be released voluntarily by the process.

4. Which of the following strategies can help prevent deadlock?

- A) Allowing circular wait
- B) Imposing a strict order of resource allocation
- C) Allowing hold and wait
- D) Forcing processes to wait indefinitely

Answer: B) Imposing a strict order of resource allocation

5. What is the purpose of the Banker's Algorithm?

- A) To terminate processes in deadlock
- B) To evaluate resource allocation requests and ensure safe allocation
- C) To manage semaphore operations
- D) To implement mutex locking

Answer: B) To evaluate resource allocation requests and ensure safe allocation

6. Which type of semaphore can take only the values 0 or 1?

- A) Counting Semaphore
- B) Binary Semaphore
- C) Mutex
- D) Resource Semaphore

Answer: B) Binary Semaphore

7. In the producer-consumer problem, what does the 'empty' semaphore track?

- A) The number of filled slots in the buffer
- B) The number of empty slots in the buffer
- C) The total number of resources
- D) The number of processes waiting

Answer: B) The number of empty slots in the buffer

8. What distinguishes a mutex from a semaphore?

- A) A mutex can be signaled by any thread.
- B) A mutex can only be unlocked by the thread that locked it.
- C) A mutex is used for counting resources.
- D) A mutex can be used for multiple processes.

Answer: B) A mutex can only be unlocked by the thread that locked it.

9. What is the main difference between deadlock and starvation?

- A) Deadlock involves resource allocation, while starvation does not.
- B) Deadlock is a temporary state, while starvation is permanent.
- C) Deadlock involves all processes being blocked, while starvation involves a process being bypassed.
- D) Deadlock can be resolved by resource allocation, while starvation cannot.

Answer: C) Deadlock involves all processes being blocked, while starvation involves a process being bypassed.

10. Which of the following is NOT a condition for deadlock?

- A) Mutual Exclusion
- B) Hold and Wait
- C) Resource Sharing
- D) Circular Wait

Answer: C) Resource Sharing